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Modeling the effects of electrode shape and separation distance on dendrite growth in solid-state batteries

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ABSTRACT

This project aims to develop a preliminary model to understand dendrite growth in solid-state electrochemical cells by examining how current density distribution is influenced by electrode shape, separation distance, and electrolyte composition. Regions of high current density are preferential for dendrite initiation. Using COMSOL Multiphysics, we model three electrode geometries with two different electrolytes and plot the current density in a horizontal slice, observing how this distribution changes with electrode separation distance (COMSOL, Inc, Burlington, MA). We find that pointed and square electrodes produce maximum current densities approximately 300 times and 3 times higher than round electrodes, respectively. The maximum current density in round and pointed electrodes decreases with increasing electrode separation distance, due to increased electric field spreading. The pointed electrode shows a single, consistent current density maximum, making this geometry ideal for studying dendrite initiation. In contrast, square electrodes' current densities increase with distance, likely due to amplified corner effects at larger separations, which increase local maxima and reduce edge uniformity. This suggests that in commercial applications, thicker electrolytes may intensify such hotspots. Future work will use operando optical microscopy to validate these predictions against physical dendrite growth.

I. INTRODUCTION

Surging global energy demands, combined with an urgent need for fossil fuel independence and economic resilience, have made the development of advanced energy storage solutions more critical than ever. Lithium-ion batteries (LIBs) are a common mode of energy storage for portable systems requiring a high energy density; most notably, they are crucial to the electric vehicle (EV) industry. Despite widespread implementation in EVs, LIBs are limited by slow charging times, short cycle life, limited driving range, and flammability due to the liquid electrolyte. Lithium metal anodes are thought to have the potential to significantly enhance battery performance, owing to their high theoretical capacity (3860 mAh/g).[1] Implementation of a lithium metal anode in the LIB proves challenging due to its interactions with the liquid electrolyte which cause dendrite growth, the formation of a solid electrolyte interface (SEI), and the inherent flammability of the liquid electrolyte. Realization of the lithium metal anode is possible with a solid-state electrolyte (SSE). Unlike its liquid counterpart, the SSE does not require a separator layer since it inherently prevents electrical contact between the electrodes while allowing lithium ions, with their small atomic radii, to intercalate through.[1] Moreover, they are mechanically and thermally stable with a high shear modulus, lowered flammability, and wide range of operating temperatures, making them safer than their liquid counterparts.[1,2] Particularly, SSEs are compatible with high capacity lithium metal anodes, allowing energy density to be significantly improved. They also offer longer cycle life and faster charging.[1] While many materials have been used to make SSEs, argyrodite-Li_6PS_5Cl (LPSCl) is notable due to its high ionic conductivity. Its performance can be further enhanced by introducing triethylene glycol dimethacrylate (TEGDMA).[3] In-situ polymerization of TEGDMA allows it to fill pores, increasing interfacial contact and hindering dendrite growth at the lithium metal interface. It has a high glass transition temperature, making it rigid at room temperature which also limits dendrites.[2] Moreover, Poly-TEGDMA further improves ionic conductivity and reduces electronic conductivity by shielding LPSCl particles.[3]

Despite the SSEs improved compatibility with lithium metal anodes, the widespread implementation of solid-state batteries (SSBs) remains challenging due to lingering interfacial and mechanical instability and dendrite formation issues. The interface between the electrodes and SSE is critical in facilitating ion transport within the battery. Hence, this interface bond must be strong and stable. Poor interfacial contact, surface gaps, and voids can increase resistance to ion transfer and lead to uneven lithium deposition.[1] During charging, uneven lithium plating on the anode surface facilitates dendrite growth by creating regions with elevated local electronic conductivity. Dendrites are microscopic needle-like lithium structures which can penetrate the electrolyte and grow from one electrode to the other during cycling, resulting in internal short circuits and battery failure.[1] They pose a significant issue in SSBs, whose lithium anodes are more susceptible to dendrite growth.

Regions of high current density in lithium metal are preferential for dendrite initiation. Since these areas have many electrons passing through them at a given time, there is faster reduction of lithium ions to lithium metal and thus faster lithium deposition. Uneven surface morphology can

create pockets of high current density, leading to a chain reaction of lithium deposition. Therefore, understanding the distribution of current within the cell is vital to understand trends in this dendritic growth. In this study, we analyze current densities in various electrode geometries and electrolyte compositions via mathematical modeling. By altering the geometry, we aim to control the current density distribution and, thus, any dendrite growth. If we can pinpoint where dendrites are likely to form, we can implement geometrical design strategies to control their growth and reduce the risk of battery failure.

II. MATERIALS AND METHODS

A. Model Geometry and Parameter Selection

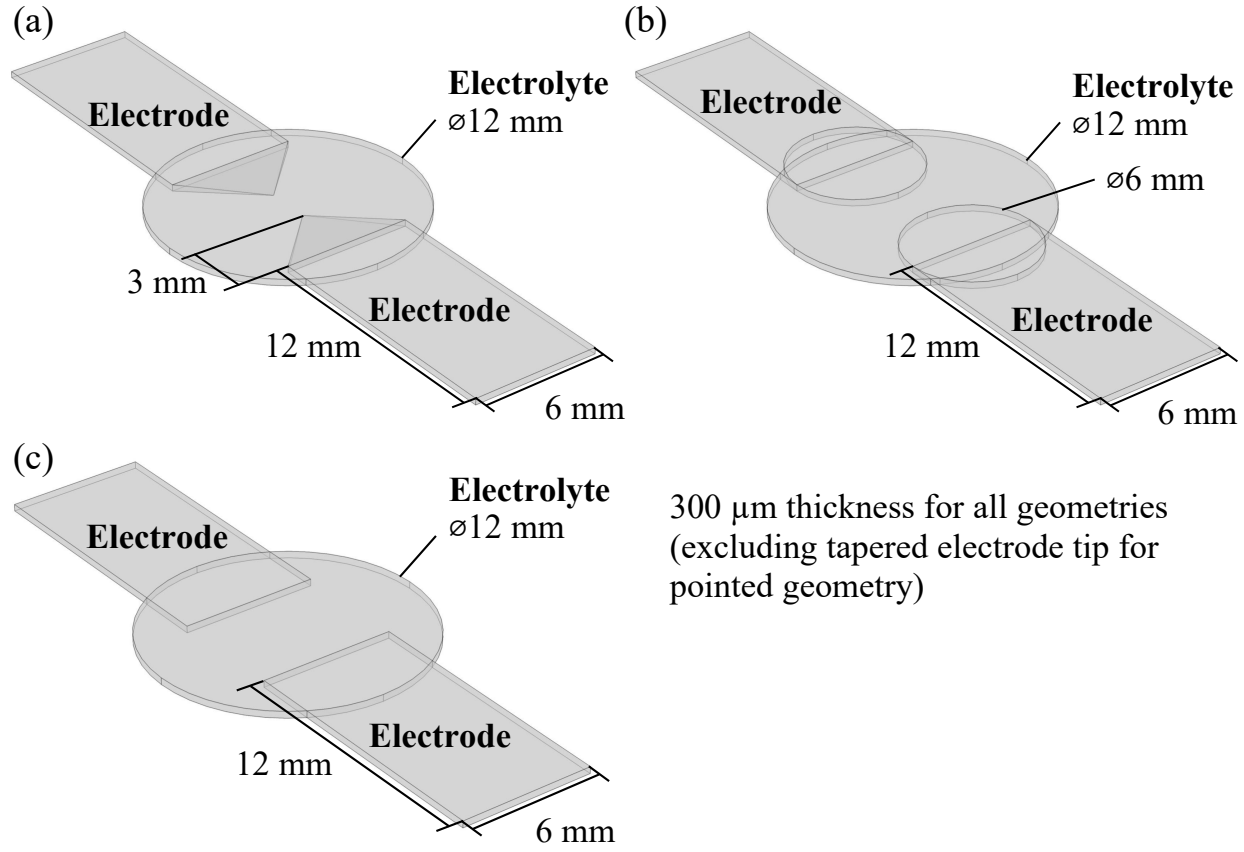


FIG. 1. Electrode geometry diagrams for electrodes with (a) round heads, (b) pointed heads, and (c) square heads.

In these simulations, three electrode geometries are modeled: round, pointed, and square. Figure 1 displays these shapes and their dimensions in greater detail. We use a symmetrical cell setup with a lithium metal anode and cathode since dendritic growth patterns can be examined from

a single electrode-electrolyte interface. Additionally, each electrode geometry is modeled with two different electrolyte materials: LPSCl and LPSCl poly-TEGDMA. The effect of altering the distance between the electrodes for each geometry and electrolyte combination is also observed. In this study, the current density surface plots for cells with electrode separation distances ranging from 0.008 mm to 5 mm are taken.

B. COMSOL Setup: Inputs, Boundary Conditions, and Meshing

This study uses a COMSOL three-dimensional simulation to model the distribution of current density within a horizontal slice of the symmetrical cell (COMSOL, Inc, Burlington, MA). Using the electric currents (ec) interface from the AC/DC module, we can evaluate the current density at many points throughout our system. The process begins by inputting the three-dimensional geometric features of the symmetrical cell system. We then select the materials to be used in our model and input the necessary material properties they are described by. Here, COMSOL uses the relative permittivity and electrical conductivity of the applied materials. Since lithium is a metal and our study is a steady-state DC model (i.e. time independent behavior is being solved for), relative permittivity plays a negligible role and is set to 1. As described by Jin et al., the ionic conductivities of LSPCl and LSPCl poly-TEGDMA are 0.243 S/m and 0.0684 S/m respectively.[2] The conductivity of lithium metal is 1.1×10^7 S/m. For the COMSOL software to fully evaluate the differential equations throughout the system, boundary conditions are necessary. In this case, we apply electric insulation around all faces excluding those where current enters and exits the cell. Those faces that do not participate in current flow require insulation to prevent current from leaking through unintended boundaries. At these faces the following equation holds:

$$\boldsymbol{J}\cdot\boldsymbol{n} = 0$$

Here, $\boldsymbol{n}$ represents a vector normal to the given surface and $\boldsymbol{J}$ represents the current density. At the uninsulated electrode faces, a fixed current density represents current entering the working electrode and exiting the counter electrode. This is modeled by applying a normal current density of +0.8333 mA/cm^2 to the working electrode (top) and -0.8333 mA/cm^2 to the counter electrode (bottom) using the following boundary conditions:

$$\boldsymbol{J}\cdot\boldsymbol{n} = +0.8333\ mA/m^2$$
$$\boldsymbol{J}\cdot\boldsymbol{n} = -0.8333\ mA/m^2$$

Finally, we employ a sufficiently fine mesh such that the software can interpolate to produce accurate results for the current density throughout the system. This requires a balance since an exceedingly fine mesh requires significant computing power while an overly coarse mesh may produce inaccurate results.

C. Governing Equations and Finite Element Analysis

To produce a continuous color surface plot of the cell's current density, COMSOL uses finite element analysis (FEA). FEA solves partial differential equations (PDEs) by breaking down a large domain into smaller, manageable elements. This is done by applying a mesh to discretize the continuous geometry. A continuous three-dimensional geometry is divided into finite tetrahedral and hexahedral elements and the vertices of each element are called nodes. COMSOL then solves the given PDE at each node and interpolates to return a continuous plot.

To plot the current density, COMSOL uses an equation for current conservation. This equation is derived from Maxwell's equations as detailed in the appendix. Since there are no internal current sources, the equation can be simplified to:

$$\nabla \cdot \boldsymbol{J} = 0 \tag{1}$$

For conductive media, the current density, $\boldsymbol{J}$, can be defined by $\boldsymbol{J} = \sigma \boldsymbol{E} + \boldsymbol{J_e}$, where σ is the conductivity, $\boldsymbol{E}$ is the electric field, and $\boldsymbol{J_e}$ represents non-Ohmic current sources within the material. Since there are no internal current sources and our system is in steady state, this equation is just the microscopic form of Ohm's law:

$$\boldsymbol{J} = \sigma \boldsymbol{E} \tag{2}$$

Lastly, we assume that our system is static, meaning fields are time-independent and there are no changing magnetic fields. Because of this, we can use the electrostatic definition of electric field:

$$\boldsymbol{E} = -\nabla V \tag{3}$$

Here, V represents the electric potential. Combining equations (1), (2), and (3) produces the following equation: $\nabla \cdot (\sigma \nabla V) = 0$. Assuming the conductivity is constant, this is simply:

$$\nabla^2 V = 0 \tag{4}$$

Using this equation, the electric potential can be solved for at all points in the geometry. For all calculations, the electric currents interface first solves for electric potential and derives electric field and current density using this information. COMSOL reuses the electrostatic definition of electric field, equation (3), to compute the electric field using the electric potential. Then, using Ohm's law, equation (2), the current density is calculated using the electric field information. Current density values are interpolated across the domain and visualized as surface plots, field plots, or streamlines. In our case, current density is visualized via surface plots in a horizontal slice at mid-height as shown in Figure 2.

III. RESULTS AND DISCUSSION

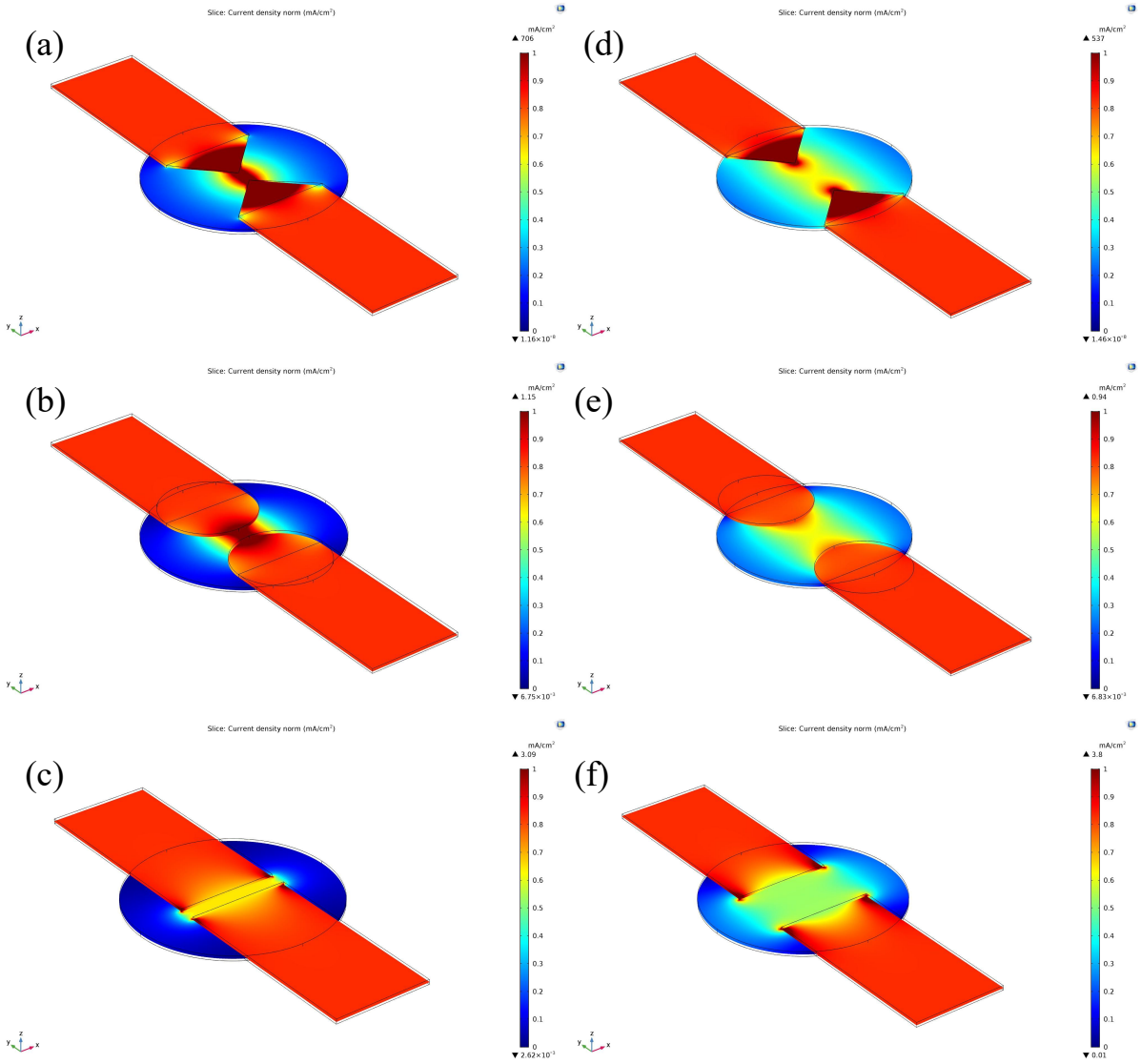


FIG. 2. Isometric view current density surface plots in cells with (a) pointed, (b) round, and (c) square geometries and 1 mm of electrode separation and (d) pointed, (e) round, and (f) square geometries with 4 mm of electrode separation. All electrolytes are solid LPSCl.

The results of the models show that geometry serves as a means of controlling current density distribution. While a round electrode head distributes current more broadly, pointed and square heads constrict current flow and facilitate high localized current densities. In fact, as can be seen in Figure 3, for separation distances 1 mm and wider, the pointed electrode setup produces maximum current densities 300 times those of the round electrode cell.

Moreover, the graphs shown in Figure 3, which compare the maximum current densities in the round and pointed electrodes to the separation distance between them, follow inverse power-law relationships. This can be explained by the fact that as electrode separation increases, electric field

spreading in the electrolyte increases, decreasing the density of field lines or the field intensity. This reduces current density based on Ohm's Law.

The current density plots provide further insight into likely sites of dendrite initiation. The location of the maximum current density for the pointed geometry is consistently at the electrode tip, making this geometry ideal for controlled studies of dendrite initiation using operando optical microscopy. In contrast, for larger separation distances in particular, the location of the maximum current density in the round electrodes is irregular, lending itself to unpredictable dendrite growth, aligning with the results of Jin et al.[2] Moreover, despite changes in the magnitude of current density, the distributions of current density are relatively independent of separation distance for both round and pointed electrodes as shown in Figures 2 and 4.

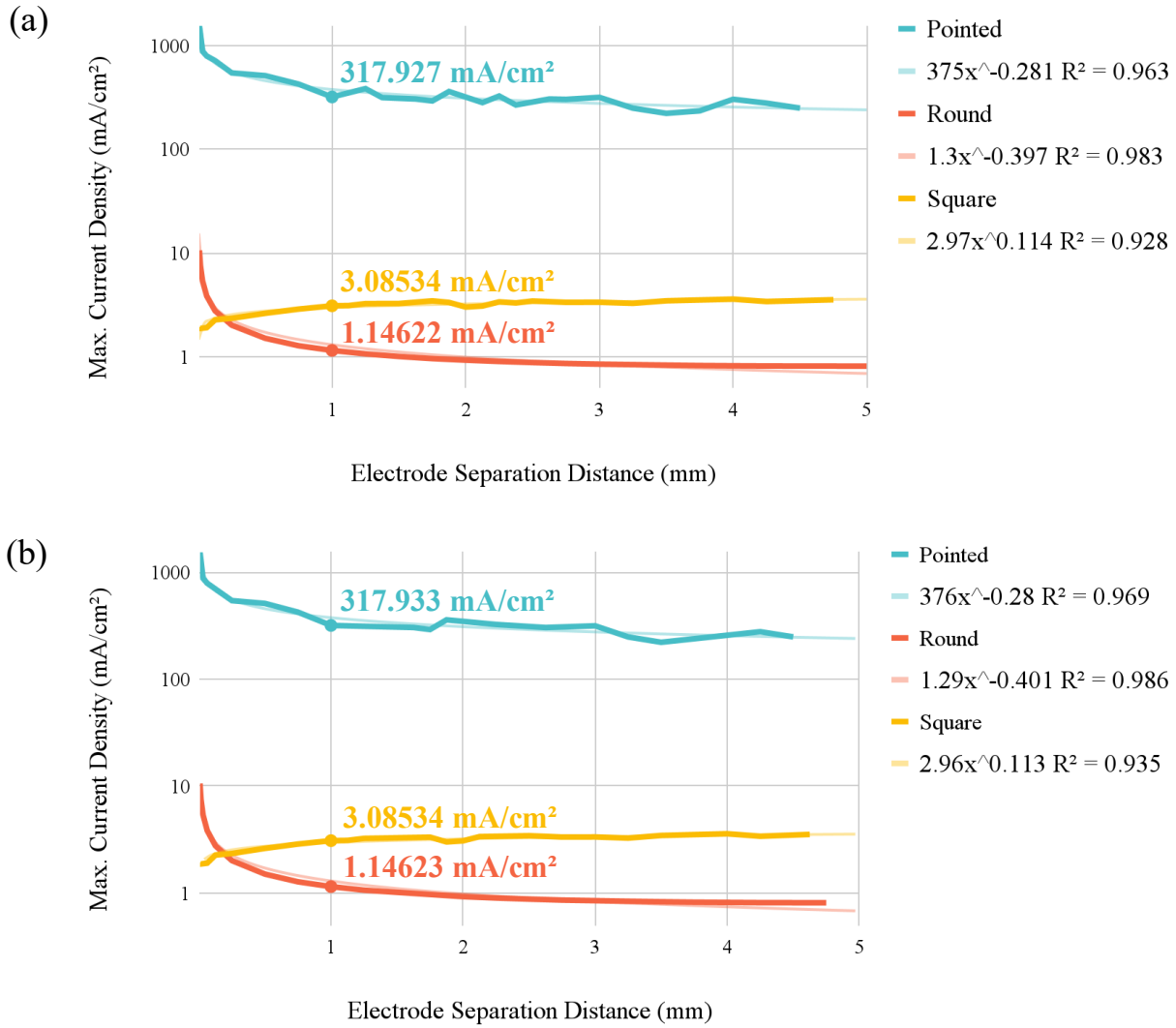


FIG. 3. Maximum current density versus electrode separation distance for pointed, round, and square electrodes and electrolytes made of (a) LPSCl and (b) LPSCl poly-TEGDMA. Maximum current density labeled for 1 mm electrode separation distance.

The results for the square electrodes, however, deviate from those of the pointed and round electrodes. Instead of an inverse power-law relationship, the maximum current density versus electrode separation distance graphs for square electrodes follow a power-law relationship with a fractional exponent (Figure 3). We hypothesize that this is because as separation increases, electric field spreading in the electrolyte increases. However, instead of lowering current densities, this field spreading promotes current flow through the corners of the electrodes, increasing current density in these regions. This, in turn, reduces the current density at the center of the electrode edge since there are no internal current sources in the model (see Figure 4c and f). So, as separation increases, the discrepancy between current density at the corners and center increases. This phenomenon is depicted in Figure 5. These results provide insight into the effects of electrolyte thickness in commercial batteries. Typically, solid electrolytes are made thick to maintain their mechanical integrity. However, a thinner electrolyte may limit current density hotspots. Though, overly thin electrolytes allow dendrites to cause battery failure more easily.

Revisiting the graphs in Figure 3, there is little difference between the current densities observed in the cells with an LPSCl electrolyte compared to those with the LPSCl poly-TEGDMA electrolyte. This indicates that there are other structural and mechanical properties of these materials that affect their susceptibility to dendrites that the model does not consider.

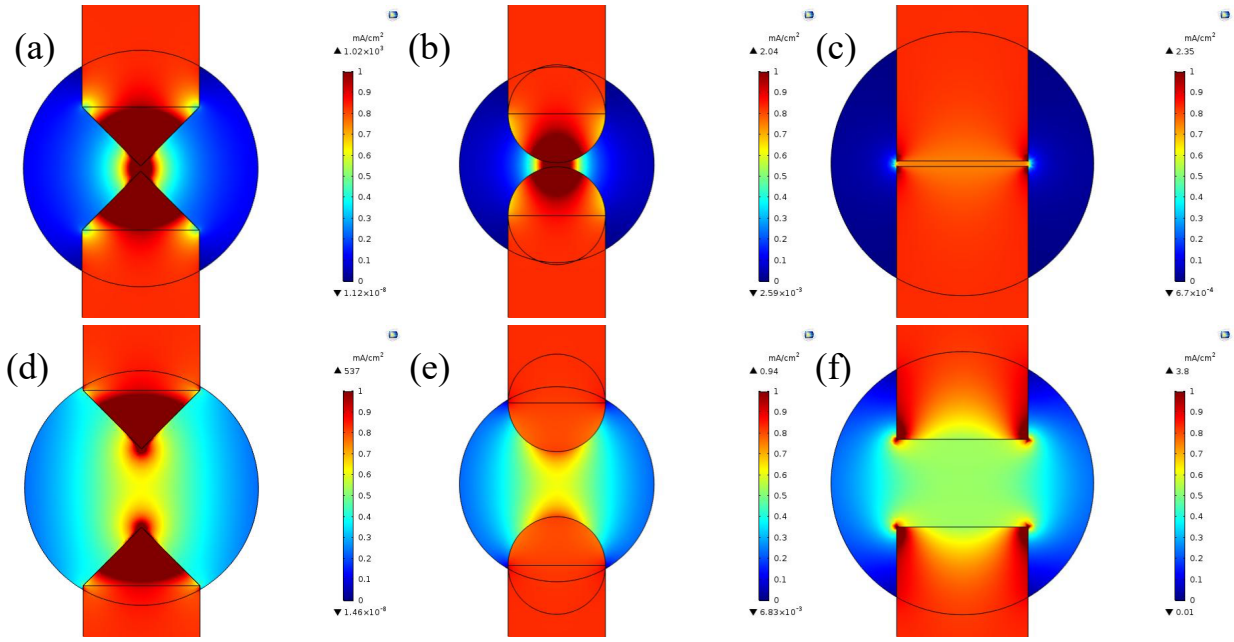


FIG. 4. Top view current density surface plots for (a) pointed, (b) round, and (c) square electrode geometries with 0.25 mm of electrode separation and (d) pointed, (e) round, and (f) square electrode geometries with 4 mm of electrode separation. All electrolytes are solid LPSCl

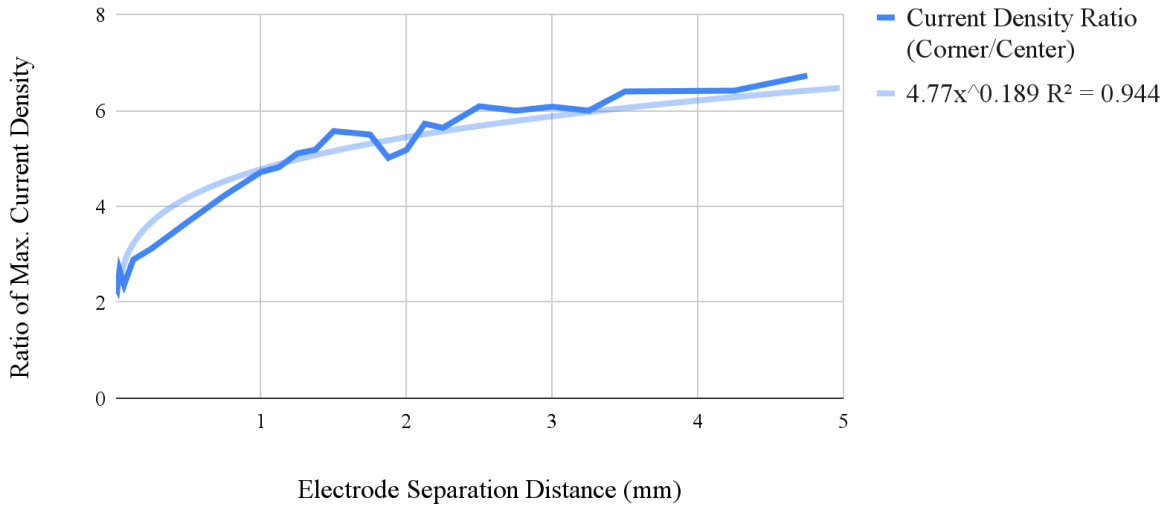


FIG. 5. Ratio of current density at corner to center of electrode edge versus electrode separation distance.

IV. CONCLUSION

In conclusion, dendrite growth may be mitigated using geometric solutions, namely, altering electrode shape. Much like Fincher et al. propose deflecting dendrites from reaching the counter electrode using engineered stress,[4] we suggest the use of controlled geometries as modes for deflecting dendrites and preventing battery failure. More broadly, our results indicate that electrode geometry plays a central role in concentrating current. Microscopic, pointed irregularities on lithium electrodes can mimic the pointed electrode model and accumulate significant current density. Therefore, precisely shaped lithium electrodes without surface roughness allow us to reap the benefits of these controlled geometries. The further development of vapor deposition technologies is essential to obtain this precision.

Possible causes of error in the study result from poor convergence of the FEA solutions, causing asymmetrical current density distributions in the symmetrical cells under certain conditions. Moreover, some models failed to compute altogether at certain electrode separation distances and geometries. This is likely due to ill-conditioned or singular matrices arising during the solution process, often caused by poor mesh quality or problem formulations that lead to unstable or non-unique solutions. Additionally, noise in the graphs seen as isolated significant outliers surrounded by accurate neighboring points, may result from local solver convergence issues or sensitivity to mesh discretization and boundary conditions. These numerical artifacts highlight the importance of mesh refinement and optimizing solver settings and inputs in future studies.

Future work will focus on verifying simulation results using the operando optical microscopy technique proposed by Jin et al., with particular emphasis on the pointed geometry owing to its consistent site of maximum current density which lends itself to easy observation. Additionally,

using the spatially resolved current density data, we aim to adapt existing dendrite growth models to our 3D setup for more accurate predictions of dendrite propagation. In particular, Liu and Lu developed a two-dimensional model coupling SEI growth and dendrite growth propagation. Their model employs geometrical calculations and the Butler-Volmer equation to analyze current density and its effect on the growth and prevention of dendrites.[5] Adapting a similar propagation model to our three-dimensional setup will allow us to create an accurate model of dendrite growth and improve our understanding of failure mechanisms in SSBs.

Lastly, this study provided much insight about current flow in square electrode geometries, which are ubiquitous in commercial batteries. Heightened current densities in the corners of square electrodes for wider separation distances highlight a need to research the effects of thinner solid electrolytes in limiting dendrite initiation. Moreover, LIB manufacturers typically employ graphite anode oversizing to provide additional lithium intercalation capacity, thereby limiting lithium plating. We hypothesize that a similar method of anode oversizing for lithium metal anodes could be used in SSBs to lower current density for the same applied current by increasing the area through which current flows. Additional modeling of cells with these widened electrodes is needed to better understand this phenomenon.

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APPENDIX

Derivation of Current Conservation

The equation for current conservation is derived by taking the divergence of both sides of the Ampère-Maxwell equation. This results in the time derivative of the divergence of the electric displacement field on one side, which can be rewritten with Gauss's Law to be the time derivative of the volumetric charge density. The derivation is shown below:

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \partial \boldsymbol{D}/\partial t$$
$$\nabla \cdot (\nabla \times \boldsymbol{H}) = \nabla \cdot (\boldsymbol{J} + \partial \boldsymbol{D}/\partial t)$$

$$0 = \nabla \cdot \boldsymbol{J} + \partial/\partial t(\nabla \cdot \boldsymbol{D})$$
$$0 = \nabla \cdot \boldsymbol{J} + \partial \rho_v/\partial t$$
$$\nabla \cdot \boldsymbol{J} = -\partial \rho_v/\partial t$$
$$\nabla \cdot \boldsymbol{J} = 0$$

Here, $\boldsymbol{D}$ is the electric displacement field, ρ_v is the volumetric charge density, and $\boldsymbol{H}$ is the magnetic field intensity.

REFERENCES

[1] A. Joshi, D.K. Mishra, R. Singh, J. Zhang, and Y. Ding, "A comprehensive review of solid-state batteries," Applied Energy **386**, 125546 (2025).

[2] X. Jin, D. Huang, Q. Miao, Z. Zhu, W. Tong, A. Videla, and G. Liu, "Operando Optical Microscopy for Visualization of Dendrite Growth in an Argyrodite LPSCl–Polymer Composite Electrolyte," ACS Energy Lett. **10**(4), 1953–1957 (2025).

[3] F. Ahmed, A. Chen, M.V.P. Altoé, and G. Liu, "Argyrodite-Li6PS5Cl/Polymer-based Highly Conductive Composite Electrolyte for All-Solid-State Batteries," ACS Appl. Energy Mater. **7**(5), 1842–1853 (2024).

[4] C.D. Fincher, C.E. Athanasiou, C. Gilgenbach, M. Wang, B.W. Sheldon, W.C. Carter, and Y.-M. Chiang, "Controlling dendrite propagation in solid-state batteries with engineered stress," Joule **6**(12), 2794–2809 (2022).

[5] G. Liu, and W. Lu, "A Model of Concurrent Lithium Dendrite Growth, SEI Growth, SEI Penetration and Regrowth," Journal of The Electrochemical Society **164**(9), A1826 (2017).